Tokens and Context

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected context record, with a weighted-information extension.

Compute capacity, choose support, preserve qualifiers and audit inclusion before independent budgeting.

Keep one habit: Keep accounting rules, source identity, exceptions, missing facts and evidence limits visible.

MAIN LESSON · SECTION 1

Recall · Retrieve earlier ideas

> Try before opening notes.

Your evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

A Distribution of Possibilities

MAIN LESSON · SECTION 12

W1 · Compute available evidence

> Preserve the declared output reserve.

Your evidence record

- Main room task: total100,instructions20,question10,output reserve20→evidence50. A=currentapproval roomB,cost30;B=currentregister roomBcapacity12,cost20;C=oldapproval roomA,cost25. Separate event picture reserves25output→evidence45;A=general20seats,cost18;B=eventcap12,cost20;C=unrelated, cost12. The repeated A/B/C labels name different cases; preserve the case heading.

- Under total 100, instructions 20, question 10 and **output** reserve 20, calculate **evidence** capacity.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Choose complete current support

> Check both pair costs and the facts supplied.

Your evidence record

- Main room task: total100,instructions20,question10,output reserve20→evidence50. A=currentapproval roomB,cost30;B=currentregister roomBcapacity12,cost20;C=oldapproval roomA,cost25. Separate event picture reserves25output→evidence45;A=general20seats,cost18;B=eventcap12,cost20;C=unrelated, cost12. The repeated A/B/C labels name different cases; preserve the case heading.

- Compare A+B,A+C,C+B for fit and support of current approval and capacity.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Preserve a qualifier

> Keep the exception and determine whether it applies.

Your evidence record

- Main room task: total100,instructions20,question10,output reserve20→evidence50. A=currentapproval roomB,cost30;B=currentregister roomBcapacity12,cost20;C=oldapproval roomA,cost25. Separate event picture reserves25output→evidence45;A=general20seats,cost18;B=eventcap12,cost20;C=unrelated, cost12. The repeated A/B/C labels name different cases; preserve the case heading.

- A passage says room B normally holds 12, except during maintenance when the approved limit is 8. Why is retaining only the first clause inadequate?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Audit inclusion versus verification

> Fitting documents do not establish faithful generated claims.

Your evidence record

- Main room task: total100,instructions20,question10,output reserve20→evidence50. A=currentapproval roomB,cost30;B=currentregister roomBcapacity12,cost20;C=oldapproval roomA,cost25. Separate event picture reserves25output→evidence45;A=general20seats,cost18;B=eventcap12,cost20;C=unrelated, cost12. The repeated A/B/C labels name different cases; preserve the case heading.

- A team says every claim is verified because all chosen documents fit in context. What further check is needed?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Budget a new evidence set independently

> Keep the first attempt before hints.

Your evidence record

- The independent task supplies source roles and costs, not actual room values. Compute fit and explain which evidence role is missing; do not invent an approved room or capacity. If D is missing, distinguish a capacity fact supported by E from the unresolved approval needed to link the response to the current event.

- Total 120, instructions 25, question 15, **output** reserve 30. Source D is current approval, cost 28; E is current capacity, cost 22; F is old approval, cost 18.

  Compute capacity, compare D+E and E+F, and state an honest fallback if D is missing.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
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> | |

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Budget new evidence independently

MAIN LESSON · SECTION 9

Project · Keep the decision and evidence

> Extend the existing project record.

Your evidence record

- Keep the learner’s question, approved source excerpt, and **output** rules inside the active context.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep first attempt, repair and a changed-case check.

Your evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Point to work that supports your judgement.

Your evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use W2: Choose complete support for Context limits and **evidence selection**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair a claim

REFERENCE · SECTION 3

Extra practice · Repair a claim

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use the worked case for Context limits and **evidence selection**. Proposed explanation: "**Token** count equals word count". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier target

REFERENCE · SECTION 5

Reference · Earlier target

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A toy system allows 12 **token**s total for **input** and **reserved output**. Instructions use 3, a source uses 7 and the planned answer needs 4. Compute the excess and propose a revision that preserves **evidence**.

▼ **Extra practice · Earlier worked case**

REFERENCE · SECTION 6

Extra practice · Earlier worked case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A toy window holds eight **token**s; prompt uses six and quotation uses three.

Does it fit? What should be preserved?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · Changed case**

REFERENCE · SECTION 7

Extra practice · Changed case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A system can use only the last two messages. Message 1 says use centimetres; messages 2 and 3 discuss plants. What instruction may be lost, and how can the next request repair this?

My practice reasoning

> **Write or draw your reasoning and evidence.**

Week 3 Test and reflect

▼ **Extra practice · A revealing test**

REFERENCE · SECTION 8

Extra practice · A revealing test

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Extra practice · Earlier transfer

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A toy system allows 12 **token**s total for **input** and **reserved output**. Instructions use 3, a source uses 7 and the planned answer needs 4. Compute the excess and propose a revision that preserves **evidence**.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Selected W4 response copy

REFERENCE · SECTION 15

Reference · Selected W4 response copy

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Total 120, instructions 25, question 15, **output** reserve 30. Source D is current approval, cost 28; E is current capacity, cost 22; F is old approval, cost 18.

Compute capacity, compare D+E and E+F, and state an honest fallback if D is missing.

My reasoning and evidence

▼ Extension · Attention as weighted information selection — W1 Identify the roles

REFERENCE · SECTION 17

Extension · Attention as weighted information selection — W1 Identify the roles

> Use the supplied values and weights beside this task.

Contract, question and response

- Attention extension reference: V1=[2,0],V2=[0,4],V3=[4,2]; supplied weights0.5,0.25,0.25. Multiply each coordinate, then sum. Excluding V3 requires renormalising permitted weights by0.75. Empty permitted mass returns no available value under this toy contract. A weight is not a source-truth or authority score.

- Explain **query**, key, **value** and weight without calling the largest weight a truth score.

My reasoning and evidence

▼ Extension · Attention as weighted information selection — W2 Compute the mixture

REFERENCE · SECTION 18

Extension · Attention as weighted information selection — W2 Compute the mixture

> Use the supplied values and weights beside this task.

Contract, question and response

- Attention extension reference: V1=[2,0],V2=[0,4],V3=[4,2]; supplied weights0.5,0.25,0.25. Multiply each coordinate, then sum. Excluding V3 requires renormalising permitted weights by0.75. Empty permitted mass returns no available value under this toy contract. A weight is not a source-truth or authority score.

- Calculate the **output** from the three supplied **value**s and weights.

> **My reasoning and evidence**

▼ **Extension · Attention as weighted information selection — W3 Mask and renormalise**

REFERENCE · SECTION 19

Extension · Attention as weighted information selection — W3 Mask and renormalise

> Use the supplied values and weights beside this task.

Contract, question and response

- Attention extension reference: V1=[2,0],V2=[0,4],V3=[4,2]; supplied weights0.5,0.25,0.25. Multiply each coordinate, then sum. Excluding V3 requires renormalising permitted weights by0.75. Empty permitted mass returns no available value under this toy contract. A weight is not a source-truth or authority score.

- Exclude V3 and calculate permitted weights and **output**. Explain the empty-**mask** case.

My reasoning and evidence

▼ Extension · Attention as weighted information selection — W4 Combine a new case independently

REFERENCE · SECTION 20

Extension · Attention as weighted information selection — W4 Combine a new case independently

Retained extension task with its own teacher key and response.

Extension question and record

- **Value**s A=[4,1],B=[0,5],C=[2,3]; weights 0.25,0.5,0.25. Compute the full **output**. **Mask** B, renormalise the remaining weights and compute the new **output**.

My reasoning and evidence

▼ Extension · Attention as weighted information selection — W5 Audit an explanation

REFERENCE · SECTION 21

Extension · Attention as weighted information selection — W5 Audit an explanation

> Use the supplied values and weights beside this task.

Contract, question and response

- Attention extension reference: V1=[2,0],V2=[0,4],V3=[4,2]; supplied weights0.5,0.25,0.25. Multiply each coordinate, then sum. Excluding V3 requires renormalising permitted weights by0.75. Empty permitted mass returns no available value under this toy contract. A weight is not a source-truth or authority score.

- A student claims the highest attention weight explains the entire model and authenticates a retrieved instruction. What additional reasoning is needed?

My reasoning and evidence

▼ Extra practice · Pictured case

REFERENCE · SECTION 23

Extra practice · Pictured case

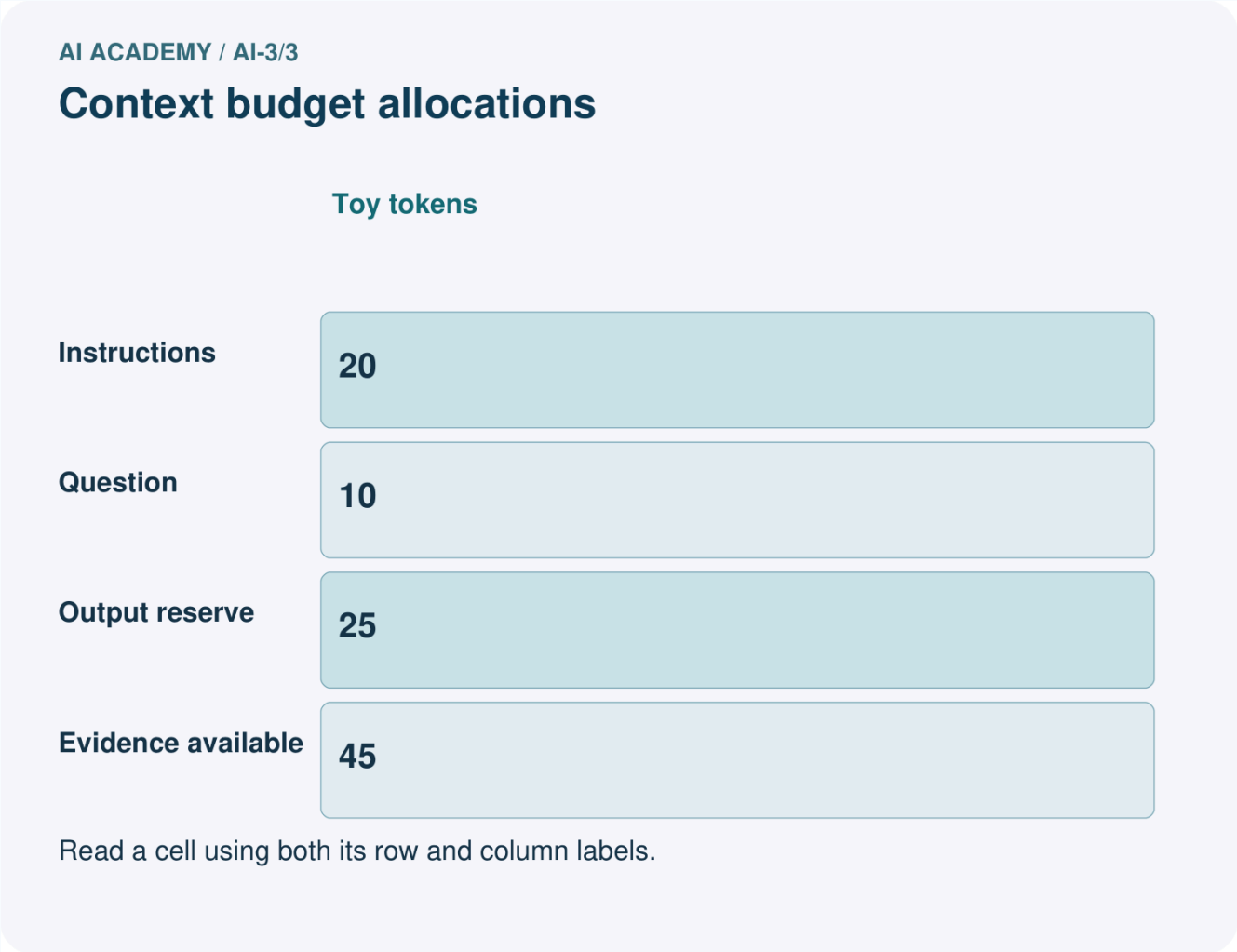
> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A fictional assistant has a small **context budget** for answering an event-capacity question.

Total budget 100 toy **token**s; instructions 20, question 10, **output** reserve 25. **Evidence** capacity is the remainder.

Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/3

Tokens and Context

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Compute budget

100-20-10-25=45 evidence tokens.

2. Select relevant pair

A+B costs 38 and preserves the capacity exception.

3. Reject incomplete shortcut

A+C costs 30 but leaves out the exception that changes the answer.

4. Answer with scope

For this event, the declared cap is 12, with both A and B retained.

CHANGE ONE THING

Increase B's cost to 30 with all other costs fixed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Can A alone justify answering 20 seats after the exception is known?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its **context budget**.

▼ **Reference · Project milestone**

REFERENCE · SECTION 24

Reference · Project milestone

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Project: Verified Study Buddy

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its **context budget**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Pictures, feedback and repair

REFERENCE · SECTION 25

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional assistant has a small **context budget** for answering an event-capacity question.
- Which labelled step matches this detail? For this event, the declared cap is 12, with both A and B retained.
- 1. Answer with scope
- 2. Select relevant pair
- 3. Reject incomplete shortcut
- 4. Compute budget
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after trying**

After your attempt, compare with the supported explanation: Answer with scope For this event, the declared cap is 12, with both A and B retained. A **context window** limits how much information is available in one model call.

Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement. Keep the relevant source relationship visible.

Toy **token** costs are supplied here; real **token** counts depend on the model's **token**izer.

Changed-case question

- Changed-case practice: Increase B's cost to 30 with all other costs fixed. Which conclusion is supported by the supplied facts?
- 1. A+B costs 48 and no longer fits the 45-**token evidence** capacity.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check changed case**

After your attempt, compare with the supported explanation: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

Repair and keep evidence

Review A Distribution of Possibilities (ai-3/2). Goal: Check whether a next-**token** table is a valid probability distribution. Short practice: Why is 0.625 not a broken probability sum?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture rather than repeating every task.

01. Tokens and Context: the case

AI ACADEMY / AI-3/3 · 01 · LOOK AT THE FACTS

Tokens and Context: the case

The situation

A fictional assistant has a small context budget for answering an event-capacity question.

Supplied evidence

Total budget 100 toy tokens; instructions 20, question 10, output reserve 25. Evidence capacity is the remainder.
Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.

By the end, I can budget context and preserve evidence needed to interpret a requested fact.

Follow the labelled picture with the matching explanation. Open the vector diagram

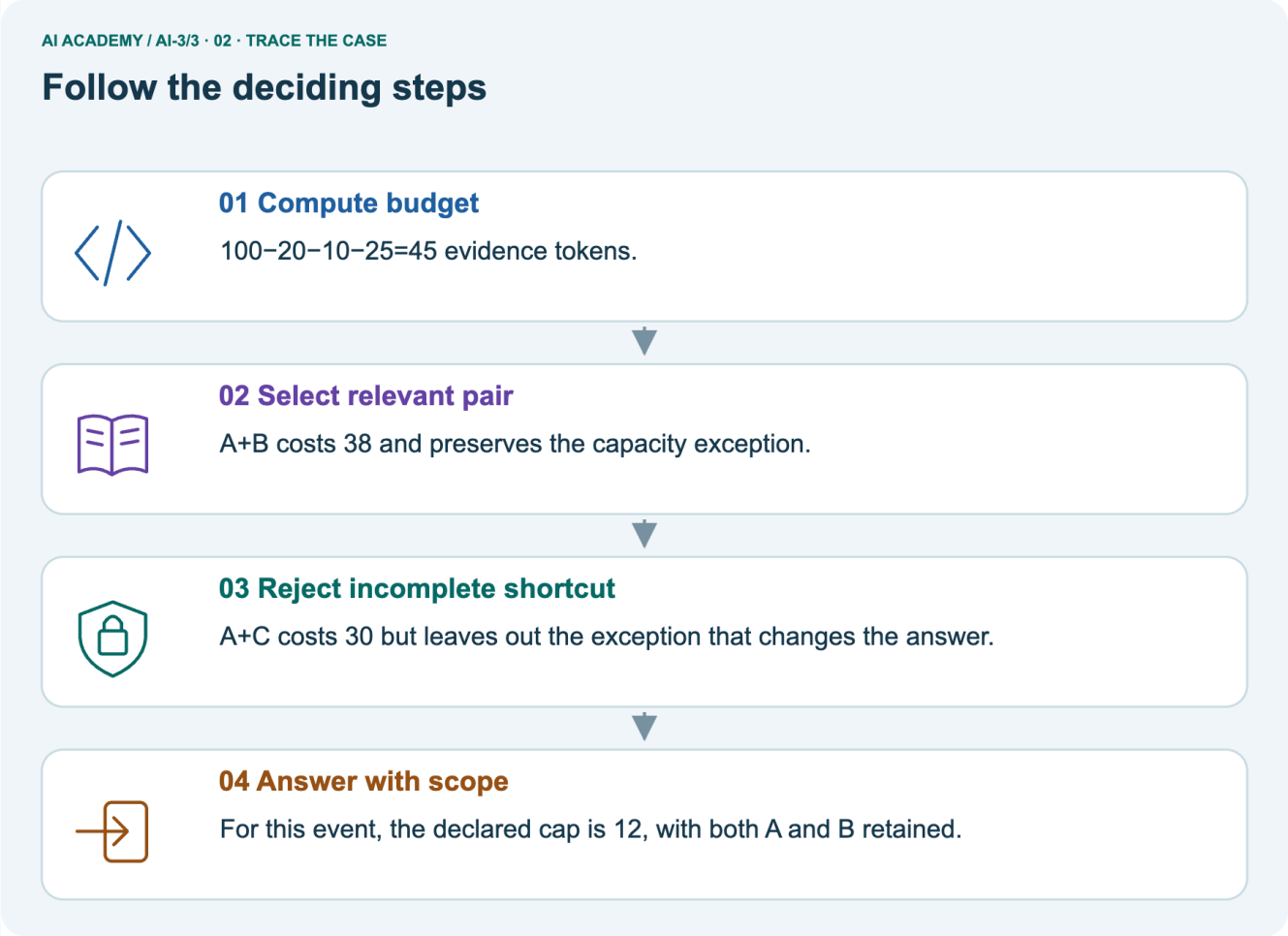
A fictional assistant has a small **context budget** for answering an event-capacity question. Supplied facts: Total budget 100 toy **token**s; instructions 20, question 10, **output** reserve 25. **Evidence** capacity is the remainder.

Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.

Point to the given fact that matters most. Explain why it matters.

> **My picture explanation, deciding evidence and one limit**

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Compute budget: 100−20−10−25=45 **evidence token**s. → Select relevant pair: A+B costs 38 and preserves the capacity exception. → Reject incomplete shortcut: A+C costs 30 but leaves out the exception that changes the answer. → Answer with scope: For this event, the declared cap is 12, with both A and B retained.

Point to each numbered stage and explain which recorded **input** it uses.

> **My picture explanation, deciding evidence and one limit**

03. Context budget allocations



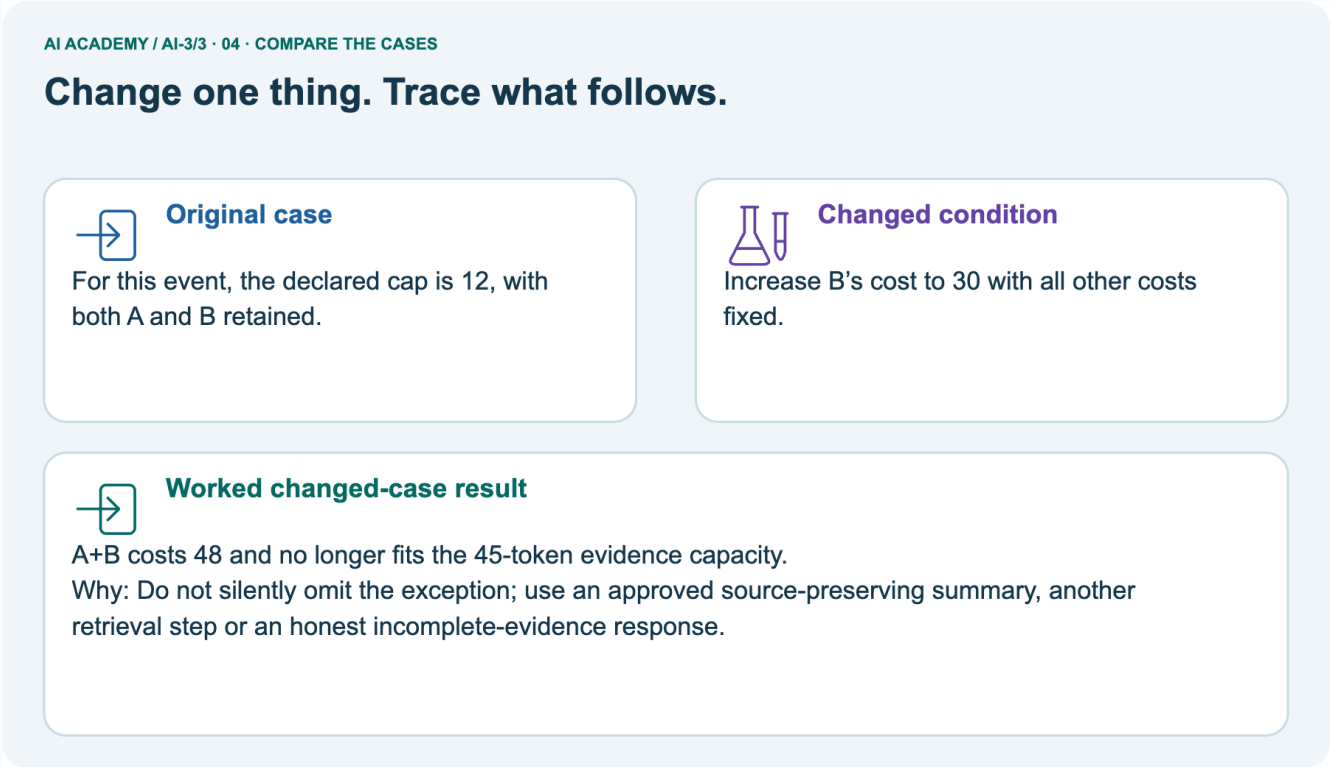
Follow the labelled picture with the matching explanation. Open the vector diagram

Context budget allocations. Instructions: Toy **token**s = 20. Question: Toy **token**s = 10. **Output** reserve: Toy **token**s = 25. **Evidence** available: Toy **token**s = 45.

Read the labels and **value**s. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: For this event, the declared cap is 12, with both A and B retained. Changed condition: Increase B's cost to 30 with all other costs fixed.

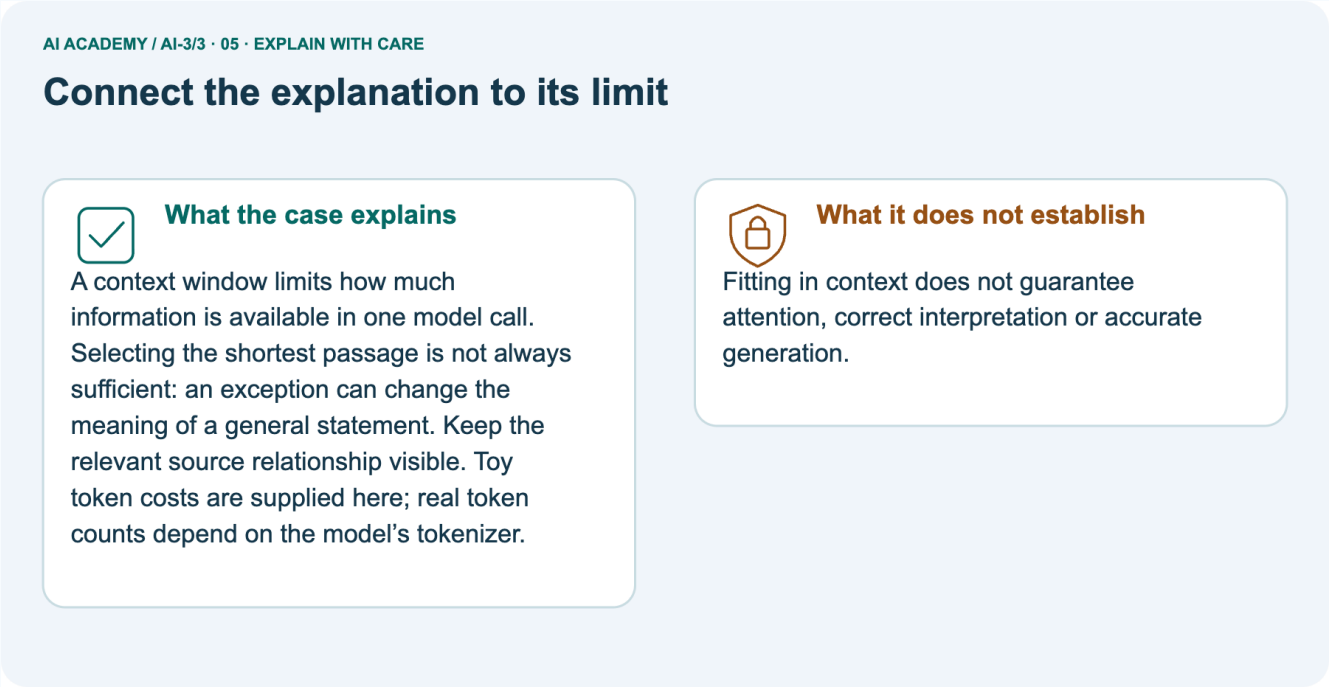
Expected result: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

Reason: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **context window** limits how much information is available in one model call. Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement.

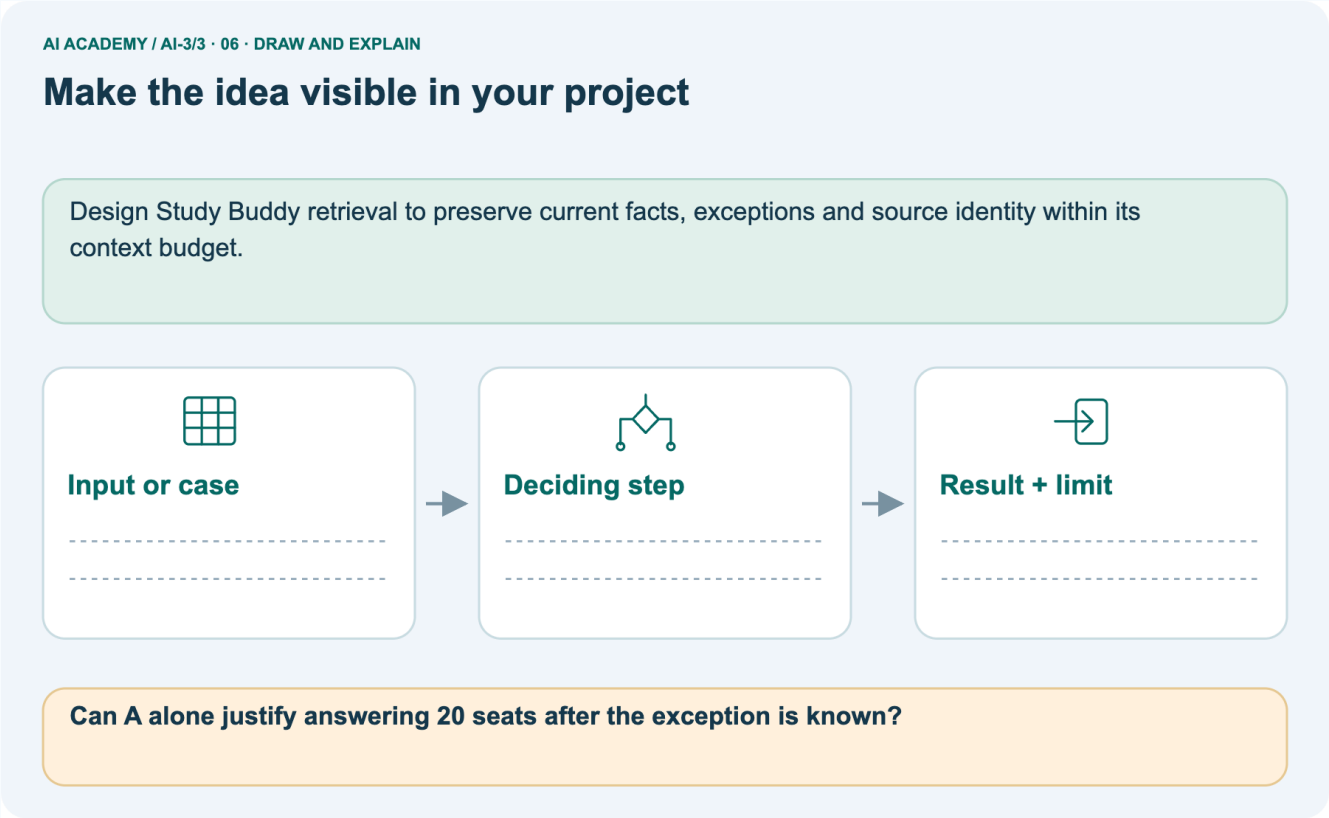
Keep the relevant source relationship visible. Toy **token** costs are supplied here; real **token** counts depend on the model's **token**izer.

Limit: **Fitting** in context does not guarantee attention, correct interpretation or accurate generation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its **context budget**. Draw your **input**, deciding step, result and limit.

Supported question: Can A alone justify answering 20 seats after the exception is known?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Tokens and Context: the case

02. Follow the deciding steps

03. Context budget allocations

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.